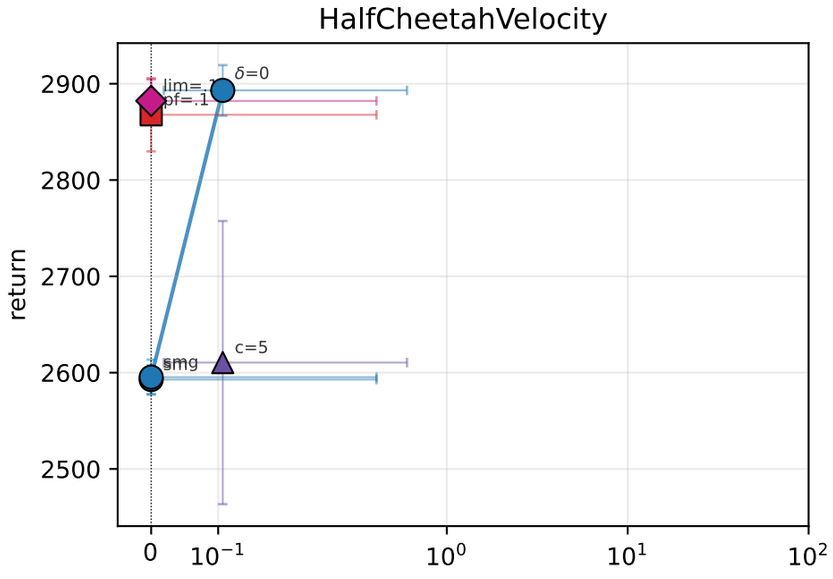
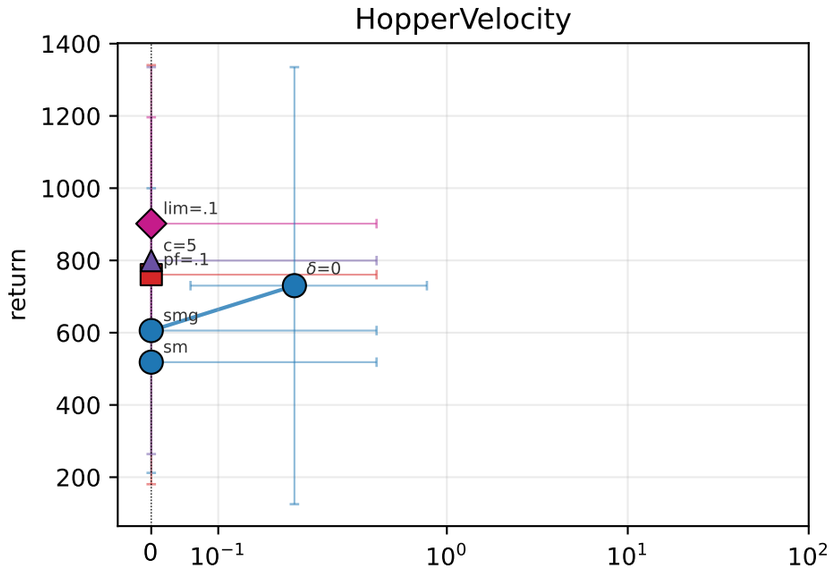
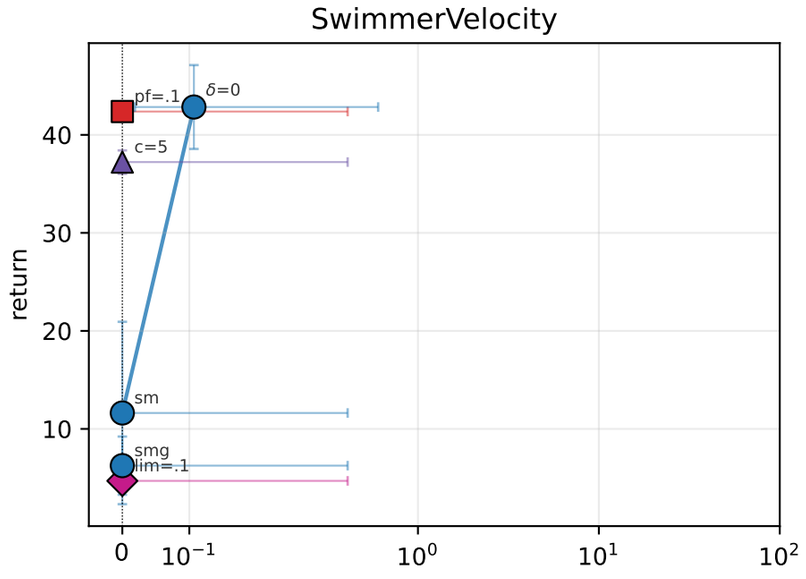
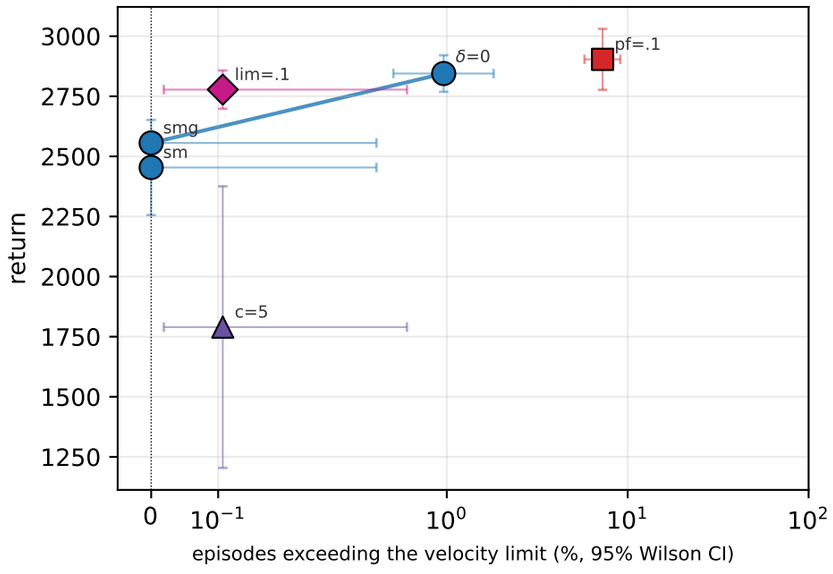
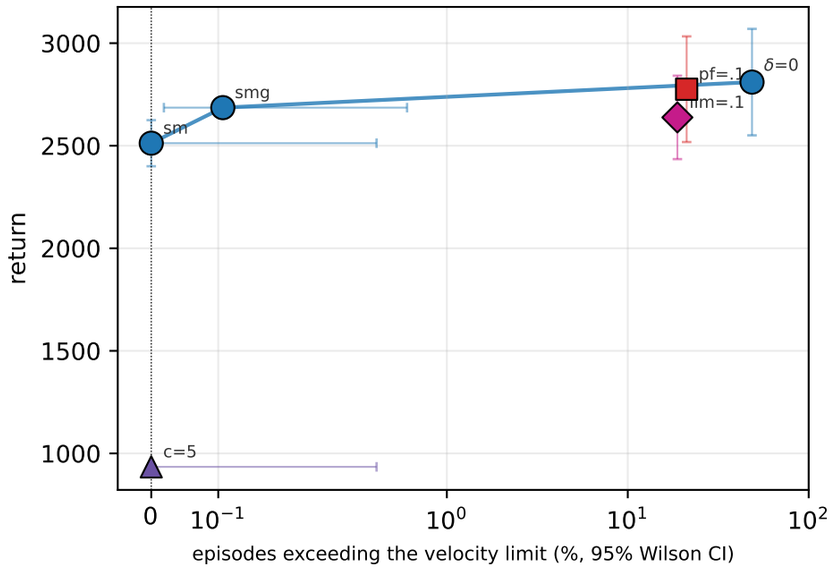
Return vs. violation rate (final checkpoint, 312 episodes/seed; single cost channel; violation = any cost in the episode)

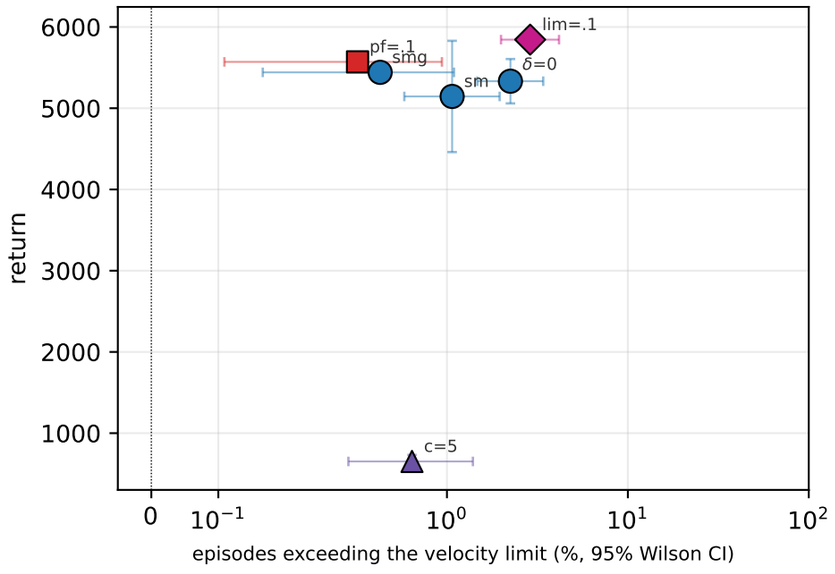












FDPI (constraint-based)
SAC-Lagrangian (penalty-based)
SafeOT-typed (ours)
SAC + fixed penalty (penalty-based)
typed Pareto frontier